NAME: _____ Geometry (FL B.E.S.T.)

9.1 Circle Vocabulary & Central Angles

A circle is the set of all points a fixed distance (the **radius**) from a center point. A **chord** connects any two points on the circle; a **diameter** is a chord through the center and is the longest possible chord, with length $2r$. A **central angle** has its vertex at the center, and its measure always equals the measure of the arc it intercepts — this is the single most important rule in circle geometry, since it converts angle problems into arc problems and back. Arcs smaller than a semicircle are **minor arcs**; arcs larger are **major arcs**; an arc cut off by a diameter is a **semicircle** ($180°$). The **Arc Addition Postulate** works just like the Segment and Angle Addition Postulates: adjacent arcs add up. Since central angles all the way around a point sum to $360°$, so do all the arcs of a circle. This worksheet uses these rules together, including algebra and real-world angle problems.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. A central angle measures $130°$. Find the minor and major arc measures.

2. $m\overset{\frown}{AB} = 52°$, $m\overset{\frown}{BC} = 88°$. Find $m\overset{\frown}{AC}$.

3. Three central angles measure $100°$, $120°$, and x. Find x.

4. A, B are diameter endpoints; $m\angle AOC = 4x$ and $m\angle BOC = 5x$. Find x and each angle.

5. A circle has radius 6.5. Find its diameter.

6. A circle has radius 8. Can it contain a chord of length 17?

7. Minor arc CD measures $95°$. Find the central angle intercepting it.

8. Central angle $\angle XOY = 48°$. Find the major arc XY.

9. $m\overset{\frown}{PQ} = 75°$ and $m\overset{\frown}{QR} = 55°$. Find $m\overset{\frown}{PR}$.

10. What is the measure of a semicircle's arc?

11. Four equal central angles fill a circle. Find each measure.

12. A circle has diameter 22. Find the radius.

13. Is a diameter also a chord? Explain briefly.

14. Four central angles around a point are in the ratio $2 : 3 : 4 : 6$. Find each angle.

15. A pizza is cut into 5 slices whose central angles form an arithmetic sequence with common difference $8°$ (each slice $8°$ bigger than the last). Find the smallest and largest slice angles.

✓ Answer Key

1. Minor $130°$; major $230°$
2. $140°$
3. $x = 140°$
4. $x = 20°$; angles $80°$ and $100°$
5. 13
6. No ($d = 16$)
7. $95°$
8. $312°$
9. $130°$
10. $180°$
11. $90°$
12. 11
13. Yes — it's the longest possible chord, since it also connects two points on the circle
14. $48°$, $72°$, $96°$, $144°$
15. Smallest $56°$, largest $88°$